

# Making up infant formula at a Target concentration

From an Australian or New Zealand formula label to a recipe of whole scoops and water to the nearest 5 mL.

## Key terms

|                                             |                                                                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Concentration</b>                        | Energy density of prepared formula, always in <b>kcal per 30 mL</b> .                                      |
| <b>Target volume / Target concentration</b> | The bottle you want to make.                                                                               |
| <b>Reconstitution statement</b>             | The label's "powder + water makes this much" line, e.g. "1 scoop + 50 mL water makes approximately 55 mL". |
| <b>kcal per scoop</b>                       | Energy in one level scoop of powder.                                                                       |
| <b>Displacement</b>                         | The volume (mL) one scoop of powder adds beyond the water it's mixed into.                                 |
| <b>Recipe</b>                               | A whole number of scoops plus water rounded to 5 mL.                                                       |
| <b>Actual volume / Actual concentration</b> | What the recipe really makes, after rounding.                                                              |

## Part A · Reading the label

Find: **energy per 100 mL** of prepared formula, the **Reconstitution statement**, and **grams per scoop**.

- A1** Scoops in the statement = grams of powder ÷ grams per scoop (*skip if the statement already counts scoops*)
- A2** Displacement = (prepared volume – water) ÷ scoops (*2 dp*)
- A3** kcal per scoop = energy per 100 mL × (prepared volume ÷ scoops) ÷ 100 (*1 dp*)

## Part B · Building the recipe

- 1 kcal/mL = Target concentration ÷ 30
- 2 kcal needed = Target volume × kcal/mL
- 3 Scoops = kcal needed ÷ kcal per scoop, **rounded up** to a whole scoop
- 4 Actual kcal = scoops × kcal per scoop
- 5 Total displacement = scoops × displacement
- 6 Water = Actual kcal ÷ kcal/mL – total displacement, **rounded up to the nearest 5 mL**
- 7 Actual volume = water + total displacement
- 8 Actual concentration = Actual kcal ÷ Actual volume × 30

**Check:** if the Actual concentration is **0.5 kcal/30 mL or more below** the Target concentration, double-check your inputs and working before using the recipe.

## Rounding

- Scoops: always round **up**, unless the answer is already exactly whole.
- Water: round **up** to a 5 mL mark, the smallest amount you can measure reliably on a bottle, unless it's already on one.
- Show kcal and volumes to 1 dp, displacement to 2 dp, Concentration to 2 dp. Keep the unrounded numbers in your calculator between steps.
- So the recipe always comes out at or just **below** the Target concentration, and at or just **above** the Target volume.

## Worked example: NAN OPTIPRO 1, 150 mL at 24 kcal/30 mL

### NAN OPTIPRO 1: the label says

**Energy:** 67 kcal per 100 mL

**Reconstitution statement:** 129 g powder + 900 mL water makes 1 L

**Grams per scoop:** Average scoop weight = 4.3 g

### Part A • Reading the label

|           | Step                    | Working                             | Answer                   |
|-----------|-------------------------|-------------------------------------|--------------------------|
| <b>A1</b> | Scoops in the statement | $129 \div 4.3$                      | <b>30</b> scoops         |
| <b>A2</b> | Displacement            | $(1000 - 900) \div 30$              | <b>3.33</b> mL per scoop |
| <b>A3</b> | kcal per scoop          | $67 \times (1000 \div 30) \div 100$ | <b>22.3</b> kcal         |

### Part B • Building the recipe

|          | Step                 | Working                                            | Answer                  |
|----------|----------------------|----------------------------------------------------|-------------------------|
| <b>1</b> | kcal/mL              | $24 \div 30$                                       | <b>0.8</b>              |
| <b>2</b> | kcal needed          | $150 \times 0.8$                                   | <b>120</b> kcal         |
| <b>3</b> | Scoops               | $120 \div 22.3 = 5.38$ , round up                  | <b>6</b> scoops         |
| <b>4</b> | Actual kcal          | $6 \times 22.3$                                    | <b>133.8</b> kcal       |
| <b>5</b> | Total displacement   | $6 \times 3.33$                                    | <b>20.0</b> mL          |
| <b>6</b> | Water                | $133.8 \div 0.8 - 20.0 = 147.3$ , round up to 5 mL | <b>150</b> mL           |
| <b>7</b> | Actual volume        | $150 + 20.0$                                       | <b>170.0</b> mL         |
| <b>8</b> | Actual concentration | $133.8 \div 170.0 \times 30$                       | <b>23.61</b> kcal/30 mL |

**Recipe: 6 scoops + 150 mL water**

**Makes:** 170.0 mL at 23.61 kcal/30 mL

**0.5 kcal/30 mL or more under target? No**

### Common mistakes

- Making the water equal to the Target volume. The powder takes up room too: subtract the total displacement.
- Rounding scoops to the nearest whole scoop instead of up.
- Forgetting to round the water up to a 5 mL mark.
- Dividing by grams per scoop when the Reconstitution statement already counts scoops.
- Mixing up per 100 mL (on the label) with per 30 mL (Concentration).
- Leaving out the check in step 8 when the bottle is small and the scoop is large.